Project: Worked examples
Title: U-values and room heat loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

U-values of built-up elements from their layers and surface resistances, and the steady-state design heat loss of one room: fabric loss through each element to its adjacent temperature, a thermal-bridge allowance, ventilation loss (through the outside temperature or an MVHR supply temperature), the room load for sizing emitters, the heat loss coefficient and the loss per floor area; optionally checked against target U-values.

WHAT IT CALCULATES	RELATION
Element U-value	$U = 1/(R_{si} + \Sigma\, d/\lambda + R_g + R_{se})$, or the user's U override for ground floors and glazing
Element fabric loss	$\Phi_i = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$, $\theta_{adj} = \theta_e$ where the element faces outside
Thermal bridges	$\Phi_{tb} = y_{tb} \cdot \Sigma(A \cdot f)_ext \cdot (\theta_{int} - \theta_e)$ on the elements facing outside
Fabric total	$\Phi_T = \Sigma\Phi_i + \Phi_{tb}$
Ventilation flow	$q_v = n \cdot V$
Ventilation loss	$\Phi_V = 0.34 \cdot q_v \cdot (\theta_{int} - \theta_e)$, or $(\theta_{int} - \theta_{supply})$ through an MVHR supply, plus $0.34 \cdot n_{inf} \cdot V \cdot (\theta_{int} - \theta_e)$ infiltration with MVHR
Room load	$\Phi_{HL} = \Phi_T + \Phi_V$ (CIBSE DHDG method: × (1 + additional allowance))
Heat loss coefficient	$HLC = \Phi_{HL}/(\theta_{int} - \theta_e)$; loss per floor area Φ_{HL}/A_{floor}
U-value check	U against the target U of each element (the global target applies to built-up elements facing outside only; an element's own target applies wherever it faces)

Basis: Building components — thermal resistance and thermal transmittance[1](homogeneous layers and surface resistances (clause numbers to be checked against the owner's copy)); Heating systems in buildings — design heat load: transmission heat loss[2](cl. 6.3.2 (clause number to be checked against the owner's copy)); Heating systems in buildings — design heat load: ventilation heat loss[2](cl. 6.3.3 (clause number to be checked against the owner's copy)); Room heat loss and the additional allowance[3](2026 ed., reduced method (name and edition unverified; to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Calculation method (changes the citations; the DHDG option adds the allowance below)	BS EN 12831-1 elemental		assumption: Elemental method; the design-team default.	any
θ_{int}	Internal design temperature	21	°C	spec: SPEC-A-02 §3.1: living-room design temperature	16 … 24 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
θ_e	External design temperature (location record)	-3	°C	assumption: Illustrative location design temperature; take the project's own from its design-temperature record.	-15 … 5 °C
layers	Construction layers (several elements share the table, keyed by element name)	8 row(s)		drawing: A-210 rev C	any
elements	Building elements bounding the room	5 row(s)		drawing: A-101 rev C	any
y_{tb}	Thermal-bridge allowance on the area facing outside	0.1	W/(m²·K)	assumption: Illustrative thermal-bridge allowance; replace with the project's bridging record.	0 … 0.3 W/(m²·K)
U_{target}	Target U for built-up elements without their own target	0.3	W/(m²·K)	spec: SPEC-A-02 §4.2: external wall U-value target	0.05 … 8 W/(m²·K)
n	Air change rate	0.8	1/h	assumption: Natural ventilation allowance for a living room.	0.1 … 3 1/h
V	Room volume	38.88	m³	drawing: A-101 rev C	5 … 1000 m³
A_{floor}	Floor area, for the loss per floor area	16.2	m²	drawing: A-101 rev C	1 … 200 m²
f_{add}	Additional allowance on the room load (CIBSE DHDG method only)	0	%	default	0 … 20 %

TABLE ROW	VALUES
layers[1]	element = Cavity wall; name = Outer brick leaf; d = 102.5 mm; lambda = 0.84 W/(m*K)
layers[2]	element = Cavity wall; name = Insulation; d = 100 mm; lambda = 0.03 W/(m*K)
layers[3]	element = Cavity wall; name = Inner block leaf; d = 100 mm; lambda = 0.2 W/(m*K)
layers[4]	element = Cavity wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m*K)
layers[5]	element = Party wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m*K)
layers[6]	element = Party wall; name = Block leaf 1; d = 100 mm; lambda = 0.6 W/(m*K)
layers[7]	element = Party wall; name = Block leaf 2; d = 100 mm; lambda = 0.6 W/(m*K)
layers[8]	element = Party wall; name = Plaster; d = 13 mm; lambda = 0.4 W/(m*K)
elements[1]	name = Front wall; construction = Cavity wall; area = 8.7 m^2
elements[2]	name = Window; area = 2.1 m^2; U_user = 1.4 W/(m^2*K)

TABLE ROW	VALUES
elements[3]	name = Gable wall; construction = Cavity wall; area = 8.64 m^2
elements[4]	name = Party wall; area = 8.64 m^2; theta_adj = 18 degC; Rg = 0.18 m^2*K/W; U_max = 1.2 W/(m^2*K)
elements[5]	name = Ground floor; area = 16.2 m^2; U_user = 0.25 W/(m^2*K)

3. Assumptions

A1. Steady state: every temperature is constant, with no solar or internal gains, no thermal mass and no dynamic simulation. (Design heat loss, not an annual energy estimate.)
A2. Layers are homogeneous and in series (one-dimensional flow); no bridged layers (timber studs, mortar joints, fixings) are corrected for: a thermal-bridge allowance y_{tb} is added instead. (BS EN ISO 6946 homogeneous-layer calculation.)
A3. A blank Rsi is the conventional BS EN ISO 6946 value for the element's heat-flow direction (0.13 m²K/W horizontal, 0.10 upward, 0.17 downward) and a blank Rse is 0.04 m²K/W for an element facing outside (no adjacent temperature, or one at or below θ_e), or the same as Rsi where it faces another space. (Small conventional BS EN ISO 6946 set (three values), not a table of design data.)
A4. The global target U applies to built-up elements facing outside only: an internal partition or floor has no outside target, so it is checked only against a target entered on its own row. (Scope of the target check.)
A5. Ground-floor U-values and glazing U-values are entered by the user (override); the calculator does not model ground heat transfer. (Stated limit.)
A6. The air constant 0.34 Wh/(m³·K) is the conventional rounded $\rho \cdot c_p$ of air (about 1.2 kg/m³ × 1005 J/(kg·K) ÷ 3600); it is derived, not read from a table. (Derived.)
A7. Design temperatures, conductivities, surface resistances and the thermal-bridge allowance are user records from the user's own licensed copies or manufacturers' data; none is embedded. (No table from a standard or a manufacturer is shipped.)
A8. method (Calculation method (changes the citations; the DHDG option adds the allowance below)) is taken as BS EN 12831-1 elemental. (Elemental method; the design-team default.)
A9. θ_e (External design temperature (location record)) is taken as -3 degC. (Illustrative location design temperature; take the project's own from its design-temperature record.)
A10. y_{tb} (Thermal-bridge allowance on the area facing outside) is taken as 0.1 W/(m^2*K). (Illustrative thermal-bridge allowance; replace with the project's bridging record.)
A11. n (Air change rate) is taken as 0.8 1/h. (Natural ventilation allowance for a living room.)
A12. Conventional BS EN ISO 6946 surface resistances (R_{si} 0.13 horizontal, 0.10 upward, 0.17 downward; R_{se} 0.04 outside, or the same as R_{si} where the element faces another space; m²K/W) are used where blank, for: Front wall, Gable wall, Party wall. (Small conventional set; enter your own where the construction needs other values.)

4. Working

[8] $R_1 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Front wall: 0.13 + (0.12202 + 3.3333 + 0.5 + 0.0325) + 0 + 0.04
= 4.1579 m²·K/W
Front wall: total thermal resistance

[9] $U_1 = 1/R$ [1]
= 1 / 4.1579
= 0.24051 W/(m²·K)
Front wall: U-value

[10] $\Phi_1 = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.7 × 0.24051 × 1 × (21 − -3) °C
= 50.218 W
Front wall: fabric loss to θ_e = -3 °C

[11] U_2 = U override (user-entered)
= Window: 1.4 W/m²K
= 1.4 W/(m²·K)
Window: U-value as entered (ground floor or glazing)
[12] $\Phi_2 = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 2.1 × 1.4 × 1 × (21 - -3) °C
= 70.56 W
Window: fabric loss to θ_e = -3 °C
[13] $R_3 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Gable wall: 0.13 + (0.12202 + 3.3333 + 0.5 + 0.0325) + 0 + 0.04
= 4.1579 m²·K/W
Gable wall: total thermal resistance
[14] $U_3 = 1/R$ [1]
= 1 / 4.1579
= 0.24051 W/(m²·K)
Gable wall: U-value
[15] $\Phi_3 = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.64 × 0.24051 × 1 × (21 - -3) °C
= 49.872 W
Gable wall: fabric loss to θ_e = -3 °C
[16] $R_4 = R_{si} + \Sigma\, d/\lambda + R_g + R_{se}$ [1]
= Party wall: 0.13 + (0.0325 + 0.16667 + 0.16667 + 0.0325) + 0.18 + 0.13
= 0.83833 m²·K/W
Party wall: total thermal resistance
[17] $U_4 = 1/R$ [1]
= 1 / 0.83833
= 1.1928 W/(m²·K)
Party wall: U-value
[18] $\Phi_4 = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 8.64 × 1.1928 × 1 × (21 - 18) °C
= 30.918 W
Party wall: fabric loss to θ_{adj} = 18 °C
[19] U_5 = U override (user-entered)
= Ground floor: 0.25 W/m²K
= 0.25 W/(m²·K)
Ground floor: U-value as entered (ground floor or glazing)
[20] $\Phi_5 = A \cdot U \cdot f \cdot (\theta_{int} - \theta_{adj})$ [2](cl. 6.3.2)
= 16.2 × 0.25 × 1 × (21 - -3) °C
= 97.2 W
Ground floor: fabric loss to θ_e = -3 °C
[21] $A_{fab} = \Sigma\, A$
= 44.28 m²
= 44.28 m²
Total fabric area
[22] A_{ext} = Σ A facing outside
= 35.64 m²
= 35.64 m²
Fabric area facing outside
[23] $A_{tb} = \Sigma\, A \cdot f$ facing outside [2](cl. 6.3.2)
= 35.64 m²
= 35.64 m²
Fabric area facing outside weighted by the temperature factor (thermal-bridge allowance applies)

[24] $U_m = \Sigma\ A \cdot U\ /\ \Sigma\ A$
= 21.467 / 44.28
= 0.48479 W/(m²·K)
Area-weighted mean U-value of the fabric (without the temperature factor f)
[25] $U_{eff} = \Sigma\ A \cdot U \cdot f\ /\ \Sigma\ A$
= 21.467 / 44.28
= 0.48479 W/(m²·K)
Heat-loss weighted mean U-value of the fabric (includes the temperature factor f)
[26] $dT = \theta_{int} - \theta_e$
= 21 degC − (-3 degC)
= 24 K
Design temperature difference, inside to outside
[27] $\Phi_{el} = \Sigma\ \Phi_1$ [2](cl. 6.3.2)
= 298.77 W
= 298.77 W
Fabric loss through the elements
[28] $\Phi_{tb} = y_{tb} * A_{tb} * dT$ [2](cl. 6.3.2)
= 0.1 W/(m²·K) × 35.64 m² × 24 K
= 85.536 W
Thermal-bridge allowance
[29] $\Phi_T = \Phi_{el} + \Phi_{tb}$ [2](cl. 6.3.2)
= 298.77 W + 85.536 W
= 384.3 W
Fabric heat loss
[30] $q_v = n * V$
= 0.8 1/h × 38.88 m³
= 31.104 m³/h
Ventilation air flow
[31] $dT_v = dT$
= 24 K
= 24 K
Ventilation temperature difference, inside to outside
[32] $\Phi_V = 0.34\ Wh/(m\text{^}3*K) * q_v * dT_v$ [2](cl. 6.3.3)
= 0.34 Wh/(m^3*K) × 31.104 m³/h × 24 K
= 253.81 W
Ventilation heat loss (0.34 Wh/m³K = conventional $\rho \cdot c_p$ of air, derived)
[33] $\Phi_{HL} = \Phi_T + \Phi_V$ [2](cl. 6.3.2)
= 384.3 W + 253.81 W
= 638.11 W
Room design heat loss
[34] $HLC = \Phi_{HL}\ /\ dT$
= 638.11 W / 24 K
= 26.588 W/K
Heat loss coefficient
[36] $\Phi_A = \Phi_{HL}\ /\ A_{floor}$
= 638.11 W / 16.2 m²
= 39.39 W/m²
Room design heat loss per floor area
[37] $u_1 = U\ /\ U_{target}$
= 0.24051 / 0.3
= 0.8017
Front wall: utilisation of the target U-value

[38] $u_3 = U / U_{target}$
= 0.24051 / 0.3
= 0.8017
Gable wall: utilisation of the target U-value
[39] $u_4 = U / U_{target}$
= 1.1928 / 1.2
= 0.99404
Party wall: utilisation of the target U-value

5. Results

SYMBOL	VALUE	DESCRIPTION
R_1	4.1579 m²·K/W	Front wall: thermal resistance
U_1	0.24051 W/(m²·K)	Front wall: U-value
Φ_1	50.218 W	Front wall: fabric loss
U_2	1.4 W/(m²·K)	Window: U-value
Φ_2	70.56 W	Window: fabric loss
R_3	4.1579 m²·K/W	Gable wall: thermal resistance
U_3	0.24051 W/(m²·K)	Gable wall: U-value
Φ_3	49.872 W	Gable wall: fabric loss
R_4	0.83833 m²·K/W	Party wall: thermal resistance
U_4	1.1928 W/(m²·K)	Party wall: U-value
Φ_4	30.918 W	Party wall: fabric loss
U_5	0.25 W/(m²·K)	Ground floor: U-value
Φ_5	97.2 W	Ground floor: fabric loss
θ_{int}	21 degC	Internal design temperature
A_{fab}	44.28 m²	Total fabric area
A_{ext}	35.64 m²	Fabric area facing outside
A_{tb}	35.64 m²	Fabric area facing outside weighted by f
U_m	0.48479 W/(m²·K)	Mean fabric U-value (area-weighted)
U_{eff}	0.48479 W/(m²·K)	Heat-loss weighted fabric U-value (with f)
dT	24 K	Design temperature difference
Φ_{el}	298.77 W	Element fabric loss
Φ_{tb}	85.536 W	Thermal-bridge allowance
Φ_T	384.3 W	Fabric heat loss
q_v	31.104 m³/h	Ventilation air flow
Φ_V	253.81 W	Ventilation heat loss
Φ_{HL}	638.11 W	Room design heat loss
HLC	26.588 W/K	Heat loss coefficient
Φ_A	39.39 W/m²	Heat loss per floor area

SYMBOL	VALUE	DESCRIPTION
u_1	0.8017	Front wall: U against target
u_3	0.8017	Gable wall: U against target
u_4	0.99404	Party wall: U against target

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	U-value of Party wall against its target (1.2 W/m²K)	u_4 = 0.99404	u_4 ≤ 1	0.994	
PASS	U-value of Front wall against its target (0.3 W/m²K)	u_1 = 0.8017	u_1 ≤ 1	0.802	
PASS	U-value of Gable wall against its target (0.3 W/m²K)	u_3 = 0.8017	u_3 ≤ 1	0.802	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: θ_e (External design temperature); layers (Layer thermal conductivity); y_{tb} (Thermal-bridge allowance). Enter each as one so its source, edition and licence travel with it.
WARNING — input.f_add: f_{add} (Additional allowance on the room load (CIBSE DHDG method only)) was not entered; the template default 0 % is used.

8. Verification

Template bs.room-heat-loss@1.4.0 Content hash 3e4b1ea3ce5d1522 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.
2. BS EN 12831-1.
3. CIBSE Domestic Heating Design Guide.